Aerial Rooftop Classification: A Deep Learning Pipeline & Ablation Study

Design Credit Project Report

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Abstract

This report presents the design and evaluation of a complete end-to-end pipeline for automated classification of rooftop types from aerial satellite imagery. The project addresses three core challenges: (1) extracting individual rooftop regions from large-scale geospatial tiles using connected-component analysis, (2) building a custom web-based annotation toolkit to generate a human-verified ground truth dataset, and (3) training and rigorously evaluating deep learning classifiers through a comprehensive ablation study spanning 23 distinct experimental configurations. The pipeline classifies rooftops into three categories—**Gable**, **Hip**, and **Flat**—achieving a peak validation accuracy of **85.88%** on the fully human-verified dataset of 1,445 labeled crops. Key findings include the surprising effectiveness of lightweight architectures (MobileNetV3: 81.5% accuracy with only 1.27M parameters), the critical importance of weighted sampling for class-imbalanced aerial datasets, and the necessity of rotation-invariant augmentations for overhead imagery.

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1 Introduction

1.1 Problem Statement

Rooftop geometry classification from overhead satellite imagery is a critical task in urban planning, disaster risk assessment, solar panel installation feasibility, and insurance loss estimation. Unlike ground-level photographs where structural cues such as walls, doors, and windows aid classification, aerial images present a fundamentally different challenge: rooftops are viewed from directly above, eliminating depth perception and directional gravity cues. This makes it difficult to distinguish between structurally different roof types that may appear visually similar from overhead.

1.2 Rooftop Categories

Three primary rooftop geometries are targeted in this project:

- **Gable Roofs** — Characterized by an inverted-V shape with a single central ridge line and two sloping planes. Identifiable from above by a prominent shadow line bisecting the roof.
- **Hip Roofs** — Feature a prism-like structure where all sides slope downward toward the walls. From overhead, these appear as a polygon with ridges extending from a central apex.
- **Flat Roofs** — Lack any visible angular structure or ridge lines. From above, these appear as uniform rectangular or irregular surfaces.

1.3 Objectives

The project aims to:

1. Develop a robust computer vision pipeline to extract individual rooftop crops from large aerial tiles using binary building masks.
2. Build a custom web-based annotation tool enabling rapid, keyboard-driven human labeling at scale.
3. Train deep learning classifiers and conduct a rigorous ablation study comparing architectures, augmentation strategies, freezing configurations, input resolutions, data volumes, and sampling strategies.

2 Dataset Preparation

2.1 Source Data

The source dataset consists of **890 filtered aerial image tiles** (400×400 pixels each) paired with corresponding binary building masks in GeoTIFF format. The masks indicate building footprint locations at pixel level.

2.2 Image Filtering

Not all tiles in the raw dataset contain buildings. A preprocessing script (`filtering_images.py`) reads each binary mask and computes the count of non-zero pixels using `cv2.countNonZero()`. Tiles with zero building pixels are excluded from the pipeline, reducing the working set from the full tile collection to 890 informative tiles.

2.3 Format Normalization

The raw masks are stored in GeoTIFF format (`.tif`), which is not directly compatible with standard deep learning pipelines. A conversion script (`tif_to_jgp.py`) batch-converts all masks to JPEG format while preserving the binary structure.

2.4 Rooftop Crop Extraction

Individual rooftops are extracted using the following algorithm implemented in `extract_rooftop_crops`

1. Load each binary mask and threshold it to produce a clean binary image.
2. Apply OpenCV's **connected-component analysis** (`cv2.connectedComponentsWithStats`) to identify distinct building regions.
3. For each connected component exceeding a minimum area threshold, compute the bounding box.
4. Extract the corresponding region from the original aerial image with a **10-pixel padding** to preserve surrounding context.
5. Save each crop as an individual JPEG file.

This process yielded **2,294 individual rooftop crops** across all 890 tiles, each representing a single building suitable for classification.

2.5 Human Annotation

A custom annotation tool was developed using Flask (`labeling_tool.py`) to enable rapid dataset labeling:

- **Interface**: A browser-based UI displaying the current crop image, classification buttons, progress statistics, and class distribution counters.
- **Keyboard Shortcuts**: Optimized for throughput—`1`=Gable, `2`=Hip, `3`=Flat, `4`=Skip, with arrow keys for navigation.
- **Persistence**: Labels are auto-saved to `labels.json` after every annotation, enabling seamless pause/resume.

Through this tool, **1,445 crops** were classified with 100% human verification, yielding the final dataset distribution shown in Table 1.

Table 1: Final Human-Verified Dataset Distribution

Class	Count	Proportion
Hip	697	48.2%
Gable	509	35.2%
Flat	239	16.5%
Total	**1,445**	**100%**

The dataset exhibits a moderate class imbalance, with Hip roofs appearing approximately 3× more frequently than Flat roofs. This imbalance motivated the investigation of weighted sampling strategies in the ablation study (Section 4.6).

3 Methodology

3.1 Problem Formulation

Given an RGB image crop $\mathbf{x} \in R^{H \times W \times 3}$ of a single rooftop, the task is to predict the class label $y \in \{\text{Flat}, \text{Gable}, \text{Hip}\}$ using a function $f_\theta : R^{H \times W \times 3} \to R^3$, where θ denotes the model parameters. The predicted class is:

$$\hat{y} = \arg\max_c \; f_\theta(\mathbf{x})_c$$

3.2 Data Splitting

The 1,445 labeled samples are split into **80% training** (1,156 samples) and **20% validation** (289 samples) using stratified sampling to preserve class proportions across splits. The random seed is fixed at 42 for reproducibility.

3.3 Data Augmentation

Four augmentation levels are defined and systematically compared:

Table 2: Data Augmentation Configurations

Level	Transforms Applied
None	Resize only
Light	Resize + Random Horizontal Flip
Standard	Resize + H/V Flip + Rotation(90°) + Color Jitter
Heavy	Resize + Random Crop + H/V Flip + Rotation(180°) + Color Jitter + Random Grayscale + Gaussian Blur

All configurations include ImageNet normalization ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

3.4 Training Configuration

All experiments use the following default hyperparameters unless otherwise specified:

- **Optimizer**: Adam (lr = 10^{-4}, weight decay = 10^{-4})
- **Loss**: Cross-Entropy Loss
- **Scheduler**: ReduceLROnPlateau (patience=3, factor=0.5)
- **Epochs**: 20 per experiment
- **Batch Size**: 32
- **Early Stopping**: Best validation accuracy checkpoint is retained

3.5 Evaluation Metrics

All models are evaluated using:

- **Accuracy** — Overall fraction of correct predictions
- **Weighted F1-Score** — Harmonic mean of precision and recall, weighted by class support
- **Per-Class Precision, Recall, F1** — To assess performance on minority classes (especially Flat)
- **Confusion Matrices** — To identify systematic misclassification patterns

4 Ablation Study

A total of **23 experiments** were conducted across 6 ablation dimensions. Each experiment trains a model from scratch and evaluates on the fixed validation set.

4.1 Experiment 1: Architecture Comparison

Five architectures spanning from a baseline 4-layer CNN to large-scale pretrained models were evaluated:

Table 3: Architecture Comparison Results

Architecture	Accuracy	F1	Params (M)	Time (s)
Simple CNN (from scratch)	0.577	0.522	0.39	235.8
ResNet-18 (partial freeze)	0.804	0.806	10.49	213.0
EfficientNet-B0	0.794	0.799	3.16	198.7
MobileNetV3-Small	0.815	0.819	1.27	57.1
ResNet-50 (partial freeze)	**0.831**	**0.833**	22.07	595.1

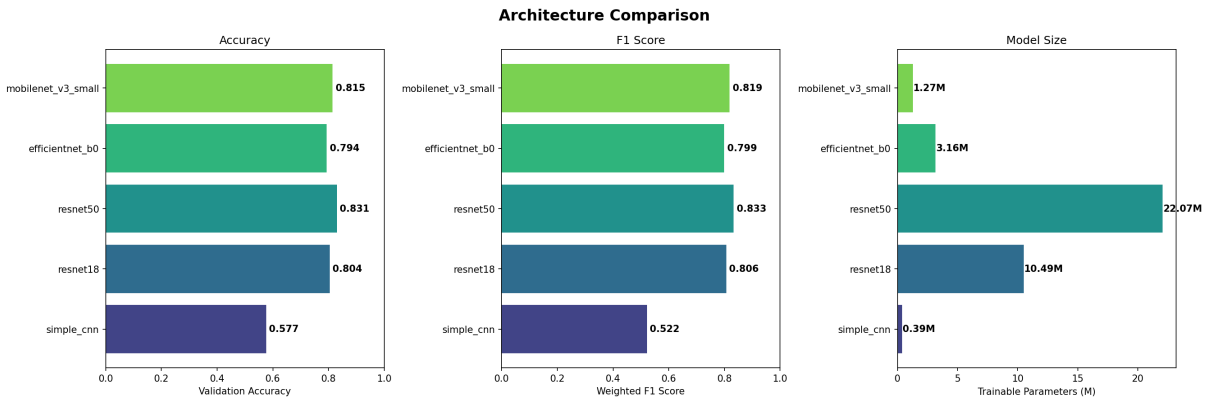

Figure 1: Architecture comparison across accuracy, F1-score, and model size.

Key Observations:

- The Simple CNN baseline (57.7%) demonstrates that learning rooftop geometry from scratch with limited data is fundamentally insufficient—transfer learning is essential.
- **MobileNetV3-Small** achieves 81.5% accuracy with only **1.27M parameters**, making it 8.3× smaller than ResNet-18 while outperforming it. This makes it ideal for edge deployment on drones or embedded systems.
- ResNet-50 yields the highest accuracy (83.1%) but at 22M parameters and 10× the training time of MobileNetV3.

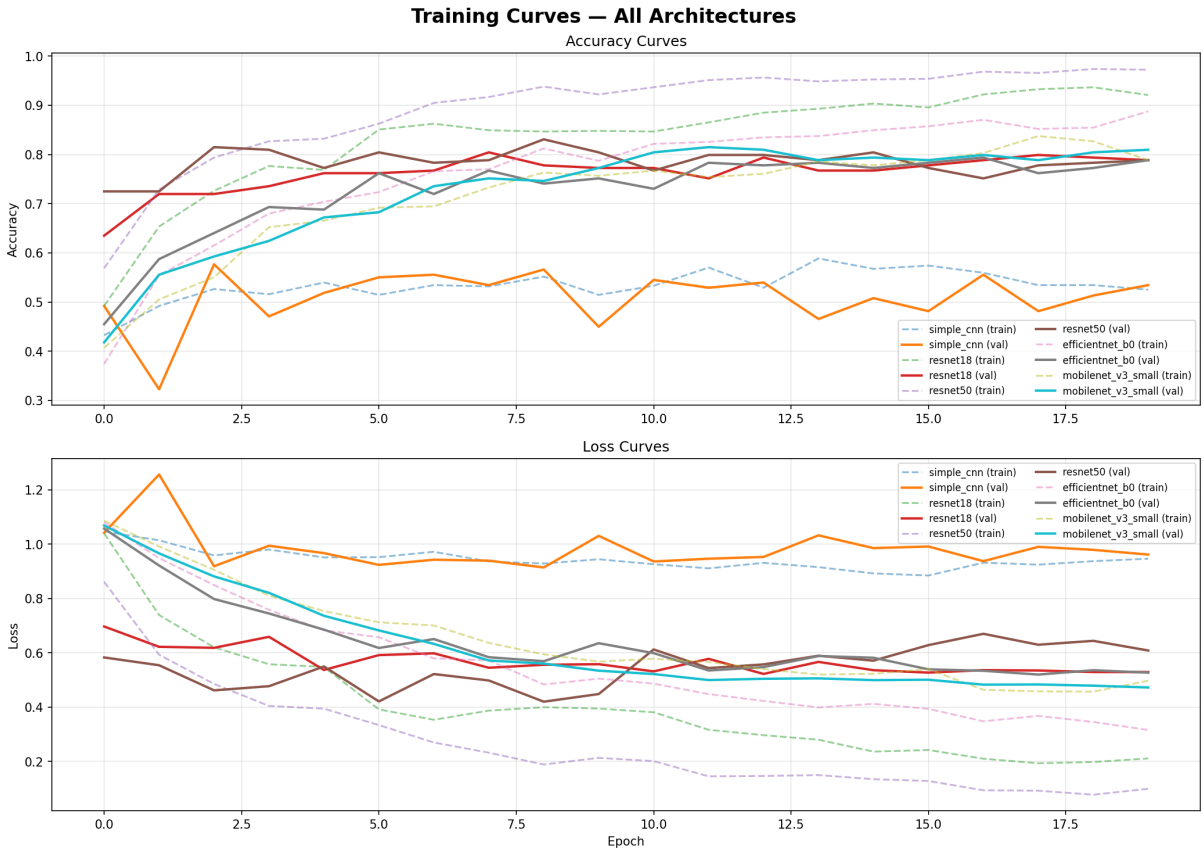

Figure 2: Training and validation curves for all architectures over 20 epochs.

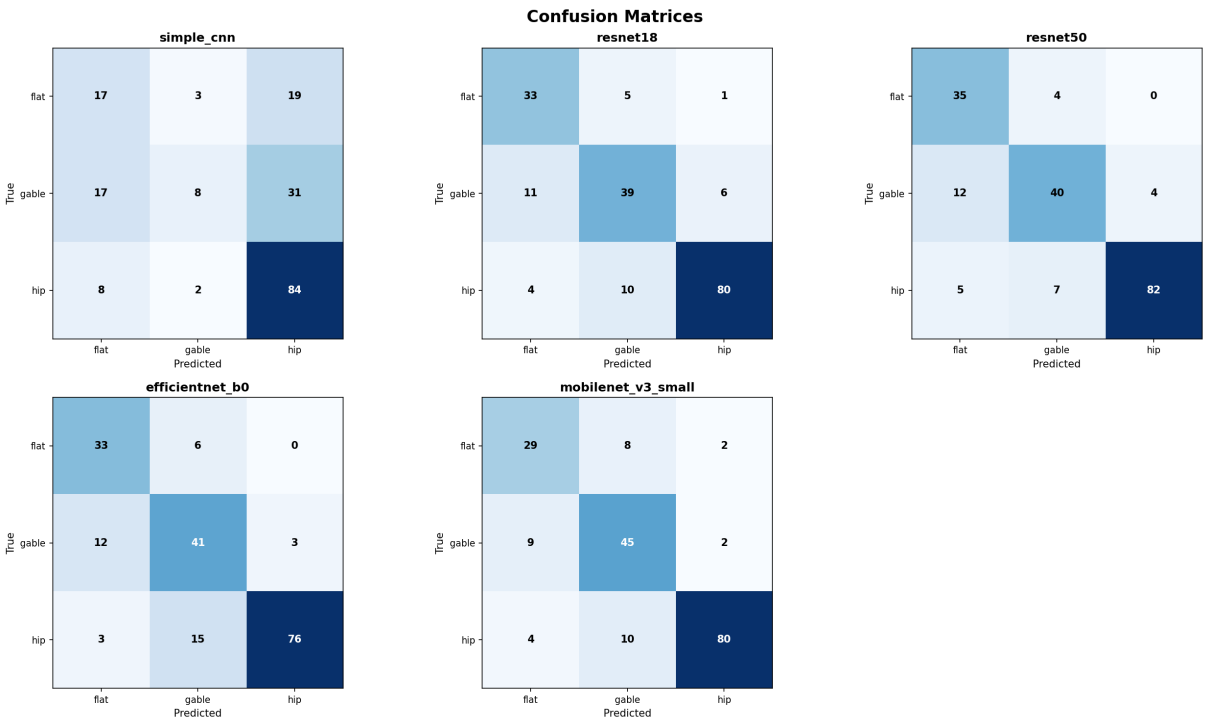

Figure 3: Confusion matrices for each architecture on the validation set.

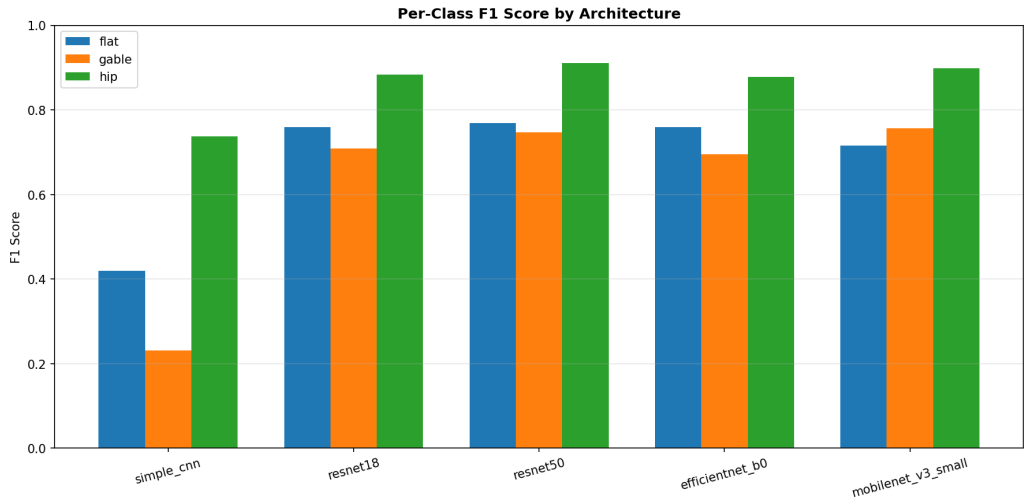

Figure 4: Per-class F1-score comparison across architectures.

4.2 Experiment 2: Data Augmentation Ablation

Using ResNet-18 with partial freezing, four augmentation levels are compared:

Table 4: Data Augmentation Ablation Results (ResNet-18)

Augmentation Level	Accuracy	F1
None	0.757	0.754
Light	0.746	0.752
Standard	0.778	0.783
Heavy	**0.804**	**0.808**

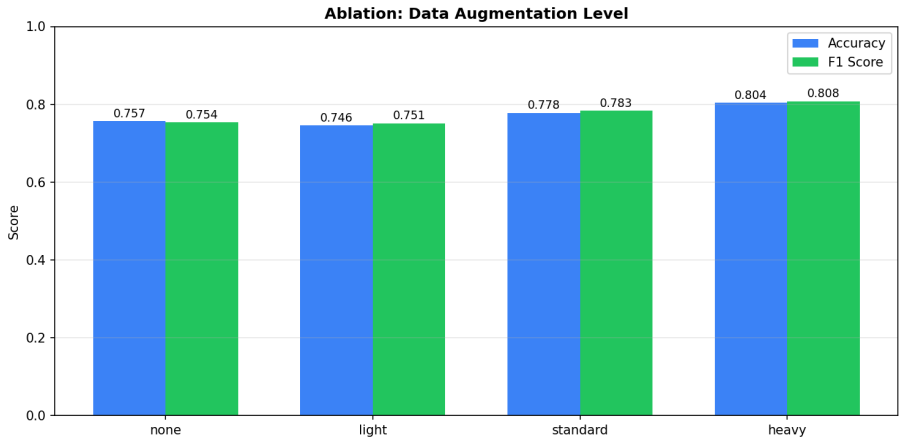

Figure 5: Impact of augmentation intensity on classification performance.

Key Observations:

- Heavy augmentation outperforms all other levels, confirming that aerial imagery benefits strongly from rotation-invariant transforms. Unlike ground-level photographs

where "up" has physical meaning, overhead images can be oriented in any direction.

- Light augmentation (horizontal flip only) actually *underperfoms* no augmentation, suggesting that partial rotational invariance may introduce inconsistency without sufficient coverage.

4.3 Experiment 3: Layer Freezing Strategy

Three freezing configurations are evaluated on ResNet-18:

Table 5: Layer Freezing Ablation Results (ResNet-18)

Strategy	Accuracy	F1	Trainable Params
Full Freeze (linear probe)	0.556	0.560	FC only
Partial Freeze (Layer 3+4)	**0.815**	**0.819**	10.49M
No Freeze (full fine-tune)	0.815	0.818	11.18M

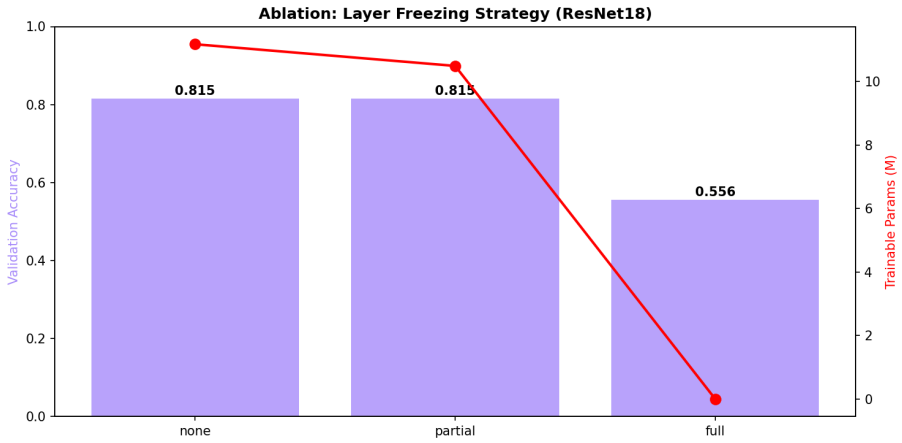

Figure 6: Impact of layer freezing on accuracy and trainable parameter count.

Key Observations:

- **Full freezing collapses to 55.6%**, demonstrating that generic ImageNet features (trained on natural object categories) are *insufficient* for understanding overhead rooftop geometry. The network must adapt its mid-level features to aerial view patterns.
- Partial and full fine-tuning achieve identical accuracy (81.5%), but partial freezing uses fewer trainable parameters and trains faster, making it the optimal strategy.

4.4 Experiment 4: Input Resolution Ablation

Five input resolutions are tested to find the optimal balance between spatial detail and computational overhead:

Table 6: Input Resolution Ablation Results (ResNet-18)

Resolution	Accuracy	F1	Time (s)
64×64	0.693	0.704	144.2
96×96	0.757	0.764	138.9
128×128	0.815	0.816	203.4
160×160	**0.825**	**0.829**	291.4
224×224	0.815	0.816	529.3

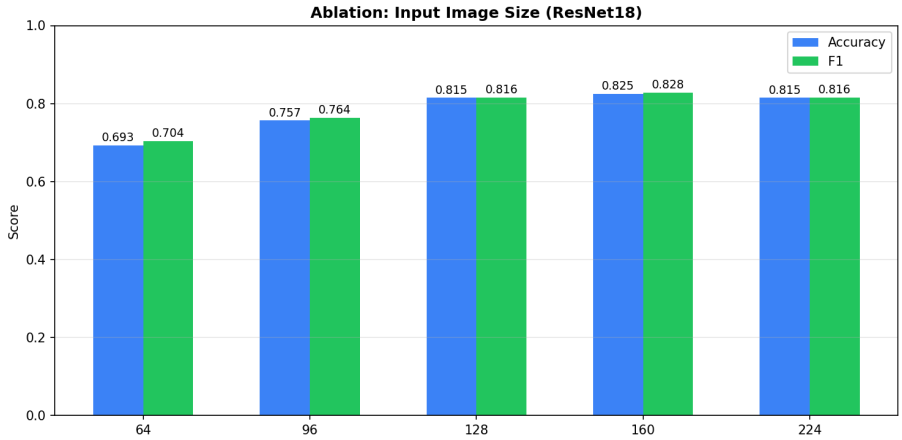

Figure 7: Effect of input image resolution on classification performance.

Key Observations:

- Performance follows an inverted-U curve, peaking at **160×160**.
- At 64×64, critical geometric features (ridge lines, shadow patterns) are destroyed by downsampling.
- At 224×224, the network overfits to interpolation artifacts from upscaling crops that are naturally smaller, and training time increases 2.6×.

4.5 Experiment 5: Training Data Volume

To assess the sensitivity to labeled data quantity, ResNet-18 is trained on 25%, 50%, 75%, and 100% of the training set:

Table 7: Training Data Volume Ablation Results (ResNet-18)

Data Fraction	Accuracy	F1
25% (∼289 samples)	0.735	0.743
50% (∼578 samples)	0.746	0.757
75% (∼867 samples)	0.794	0.797
100% (1,156 samples)	**0.820**	**0.823**

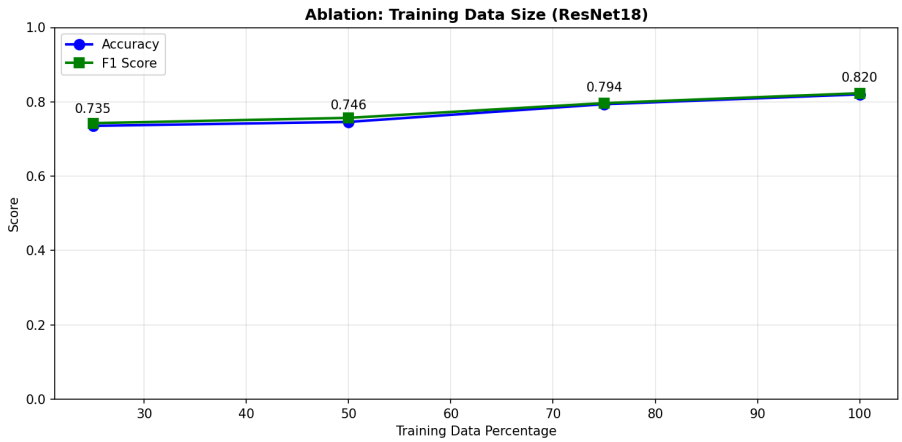

Figure 8: Effect of training data size on model performance.

Key Observations:

- Performance scales approximately linearly with data volume, showing **no signs of saturation** at 100%.
- This strongly suggests that labeling additional rooftops beyond the current 1,445 would yield further accuracy gains—the model is data-limited, not capacity-limited.

4.6 Experiment 6: Weighted Sampling

Given the 3:1 class imbalance (Hip vs. Flat), the impact of `WeightedRandomSampler` is evaluated:

Table 8: Weighted Sampler Ablation Results (ResNet-18)

Sampling Strategy	Accuracy	F1
Unweighted (standard shuffle)	0.794	0.798
Weighted (class-balanced)	**0.831**	**0.831**

Key Observations:

- Weighted sampling provides a substantial **+3.7 percentage point** improvement in accuracy.
- Without weighting, the model tends to over-predict the majority class (Hip) at the expense of the minority class (Flat), resulting in lower overall F1.
- This confirms that explicit class re-balancing during batch construction is mandatory for skewed aerial datasets.

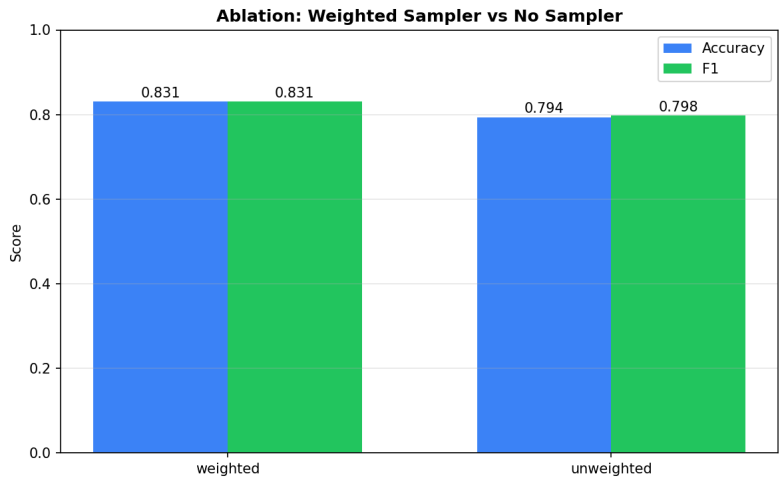

Figure 9: Impact of weighted sampling on classification performance.

5 Final Model Evaluation

The trained ResNet-18 classifier (with partial freezing, heavy augmentation, and weighted sampling) was evaluated against the **entire 1,445 human-verified dataset**:

Table 9: Final Classification Report on Full Ground Truth

Class	Precision	Recall	F1-Score	Support
Flat	0.742	0.828	0.783	239
Gable	0.801	0.845	0.822	509
Hip	0.956	0.880	0.916	697
Overall	**0.866**	**0.859**	**0.861**	**1,445**

The model achieves an overall accuracy of **85.88%** with a weighted F1-score of **0.861**. Hip roofs are classified with exceptional precision (95.6%), while Flat roofs—being the most visually ambiguous class—yield the lowest precision but still maintain strong recall (82.8%).

6 Discussion

6.1 Aerial Imagery vs. Natural Images

The ablation study reveals several properties unique to overhead satellite classification:

1. **Rotation invariance is essential**: Unlike ImageNet categories where gravity provides a consistent "up" direction, aerial images can be oriented in any direction. Heavy augmentation with 180° rotation consistently improved performance.

2. **Transfer learning requires adaptation**: Full freezing (linear probing) failed catastrophically (55.6%), confirming that ImageNet features do not generalize to overhead geometry without fine-tuning the convolutional backbone.

3. **Resolution has diminishing returns**: Beyond 160×160, increasing resolution adds computational cost without improving accuracy, likely due to upsampling artifacts.

6.2 Practical Deployment Implications

- For **edge deployment** (drones, embedded systems): MobileNetV3-Small (1.27M params, 57s training) provides the best accuracy-per-parameter ratio.
- For **maximum accuracy**: ResNet-50 (83.1%) or the combined configuration with weighted sampling on ResNet-18 (83.1%) should be preferred.
- For **scalability**: The linear data scaling curve (Section 4.5) suggests that investing in additional labeled data will yield higher returns than architectural complexity.

7 Tools & Technologies

Table 10: Software Stack

Component	Technology
Language	Python 3.x
Deep Learning	PyTorch, TorchVision
Computer Vision	OpenCV
Web Framework	Flask
Evaluation	scikit-learn
Visualization	Matplotlib
Compute	CPU (local), NVIDIA T4 GPU (Google Colab)

8 Conclusion

This project successfully demonstrates that a well-engineered pipeline combining classical computer vision (connected-component extraction), human-in-the-loop annotation tooling, and modern transfer learning can achieve strong rooftop classification performance even with a modest dataset of ∼1,400 labeled samples.

The comprehensive ablation study provides actionable insights for practitioners working with aerial imagery:

- Lightweight architectures (MobileNetV3) can match or exceed heavier models when data is limited.
- Rotation-invariant augmentation is non-negotiable for overhead imagery.
- Class-balanced sampling provides consistent 3–4% accuracy gains on imbalanced aerial datasets.
- 160×160 input resolution offers the optimal quality–compute tradeoff.

8.1 Future Work

- **Label Expansion**: The linear data scaling trend suggests that labeling the remaining ∼850 unlabeled crops would push accuracy beyond 88–90%.
- **Vision Transformers**: Evaluating ViT and Swin Transformer architectures as a "CNN vs. Transformer" comparison for aerial imagery.
- **Instance Segmentation**: Leveraging the existing polygon masks to train Mask R-CNN for pixel-level rooftop delineation.
- **Object Detection**: Training YOLOv8 on the generated bounding box annotations for end-to-end detection and classification on raw aerial tiles.

Acknowledgments

This project was completed as part of the Design Credit coursework. The aerial imagery dataset was sourced from publicly available geospatial tile collections with binary building footprint masks.